

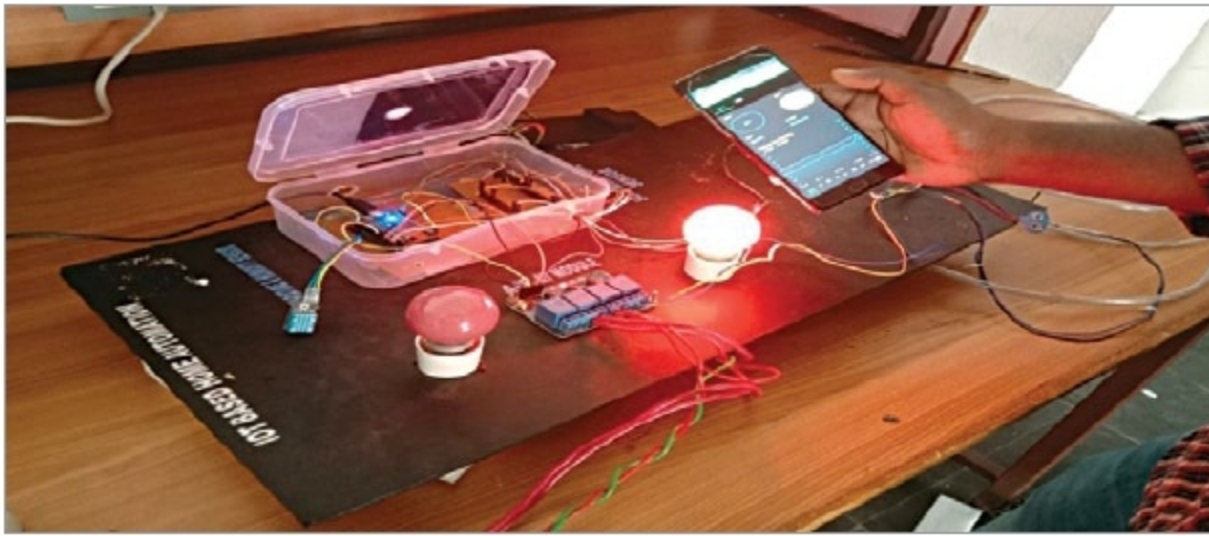


Fig. 8: Authors' prototype of the IoT-based smart home controller

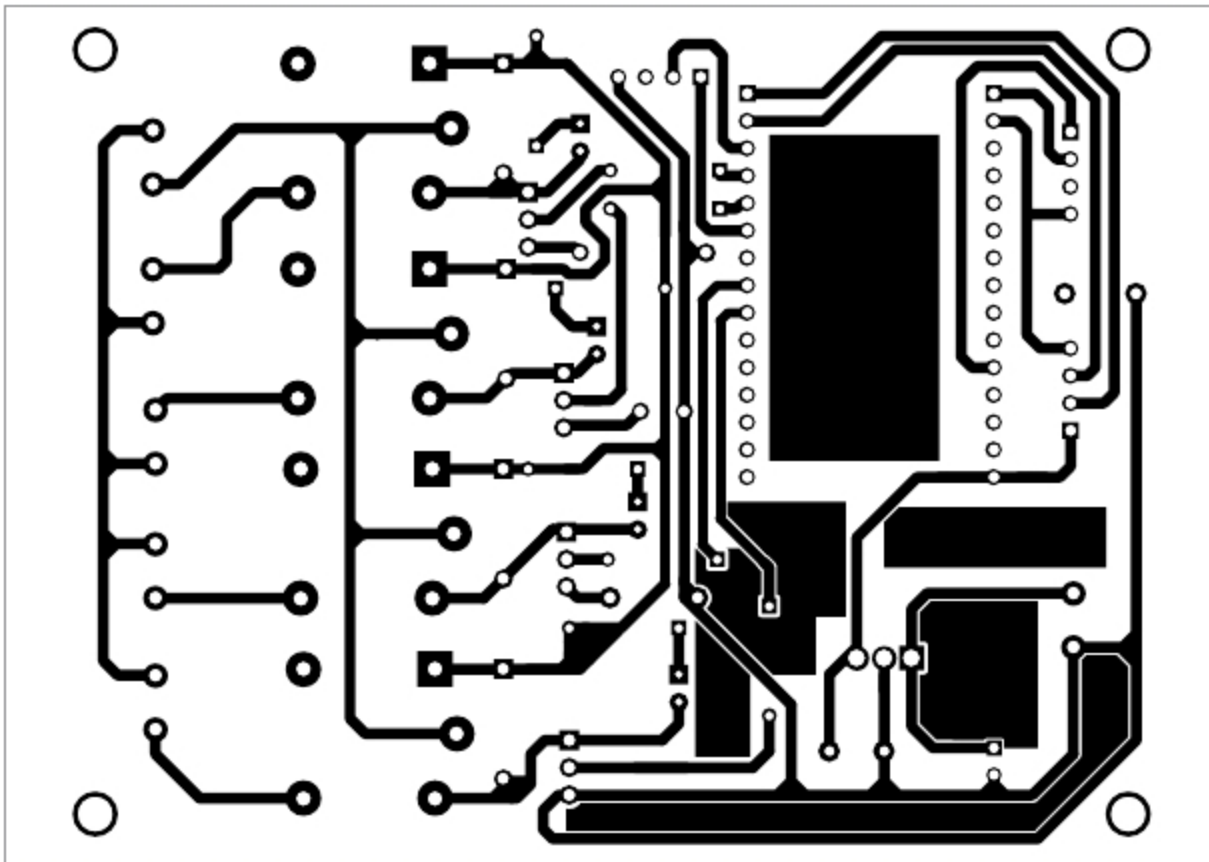


Fig. 9: Actual-size PCB layout of the IoT-based smart home controller

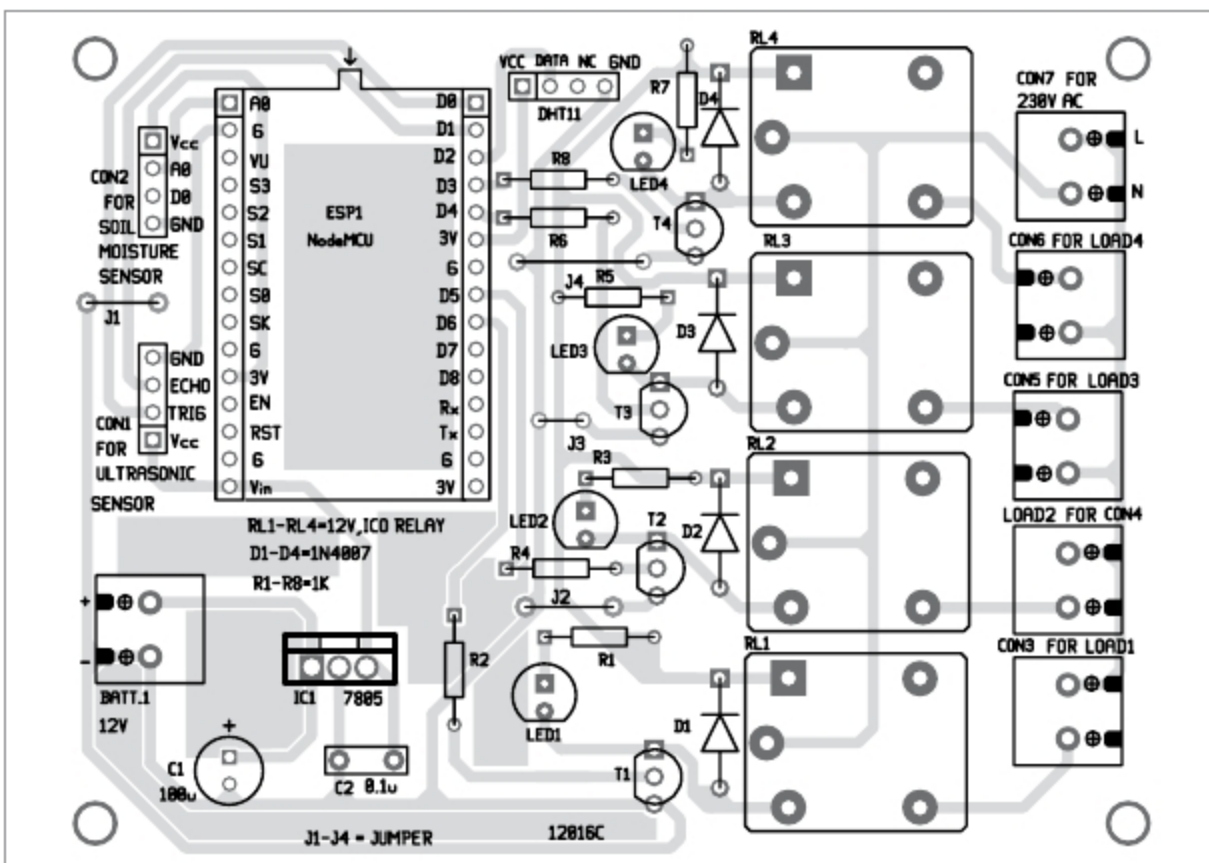


Fig. 10: Components layout for the PCB

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

ESP1	- NodeMCU Wi-Fi module
IC1	- 7805, 5V regulator
T1-T4	- BC547 npn transistor
D1-D4	- 1N4007 rectifier diode
LED1-LED4	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R8	- 1-kilo-ohm
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Capacitors:

C1	- 100 μ F, 35V electrolytic
C2	- 0.1 μ F ceramic

Miscellaneous:

CON1, CON2	- 4-pin female Berg connector
CON3-CON7	- 2-pin terminal connector
BATT.1	- 2-pin terminal connector for 12V battery
DHT11	- 4-pin female Berg connector for DHT11
RL-RL4	- 12V, single-changeover relay
	- 12V battery
	- 4-pin DHT11 temperature and humidity sensor
	- 4-pin, 5V DC soil moisture sensor
	- 4-pin ultrasonic module HC-SR04

(All these are available from kitsnspares.com)

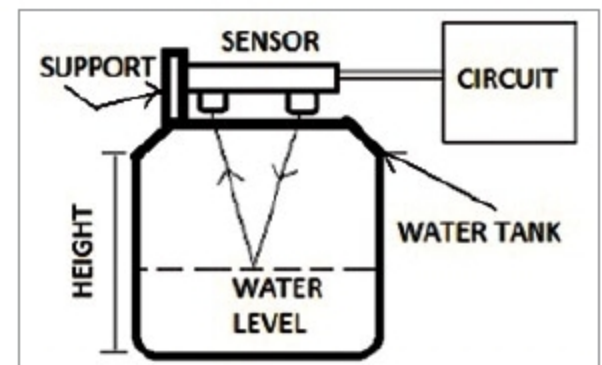


Fig. 11: Installation of ultrasonic sensor on the water tank

pump is controlled through pin D3 of NodeMCU. You can turn it on/off through the app or manually. You can monitor water level in overhead water tank from the app, and if the tank is full, you can manually turn off the motor. In case you want to operate the motor automatically, add Event in Event Manager when tank is empty or full. **EFY**



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